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一种电视页面合规性检测的方法及系统 (Chinese)

Method and system for detecting compliance of television page (English)

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Abstract (Chinese)

本发明公开了一种电视页面合规性检测的方法及系统，本发明通过构建电视页面资源标识树，将结构复杂多样的电视页面转化为具有层次的结构化模型，把电视页面受控采集、页面信息结构化建模、合规判别规则生成与治理、沙箱执行、检测报告生成等步骤抽象为可通过MCP协议调用的服务，并通过智能体实现检测任务的自动编排和判别脚本的自动生成，从而实现自动对被检测电视页面进行智能化合规性检测，继而有效解决了现有技术中无法适应复杂多样的电视页面和动态更新的监管要求等问题。

[Image Omitted]

Abstract (English)

The invention discloses a method and a system for detecting the compliance of a television page, which are characterized in that a television page resource identification tree is constructed, a television page with a complex and various structures is converted into a hierarchical structured model, the steps of controlled acquisition of the television page, structured modeling of page information, compliance judging rule generation and management, sandbox execution, detection report generation and the like are abstracted into services which can be called by an MCP protocol, and automatic arrangement of detection tasks and automatic generation of judging scripts are realized through an intelligent agent, so that the intelligent compliance detection of the detected television page is realized, and the problems that the monitoring requirements of the complex and various television pages and dynamic update cannot be adapted in the prior art are effectively solved.

[Image Omitted]

Synopsis AI Document Summarization (Default)

Summary Size: Default

Key Phrases

- judging script
- compliance detection
- script management
- rule management

In a first aspect, the present invention provides a method of television page compliance detection, the method comprising:

collecting a television page to be detected, carrying out structural analysis and element information identification on the collected television page, and extracting page information into a hierarchical and computable data model of a page-column-element structure, namely a resource information tree RIT;

Converting the management specification item into an executable judging script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script;

Executing a judging script on each resource identification tree of the structured model under the sandbox running environment, outputting a violation result, a running log and performance indexes, and carrying out violation page reproduction according to triggering;

And converting the management specification entry into an executable DSL rule by a specification analysis agent, converting the DSL rule into a discrimination script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted DSL rule and the discrimination script.

Optionally, the management specification item is converted into an executable DSL rule, the DSL rule is converted into a judging script, and the method further comprises the steps of directly calling when the required management specification item, DSL rule and judging script exist in a database, generating a new management specification item, DSL rule and judging script by the specification analysis agent when the required management specification item, DSL rule and judging script are absent in the database, and carrying out grammar and semantic verification and security and specification inspection on the generated management specification item, DSL rule and judging script and then storing the new management specification item, DSL rule and judging script in the database.

Optionally, constructing an isolated running environment for the judging script, unifying dependence, resources and random seeds, and generating a running environment number, wherein the running environment number is used for identifying a sandbox execution instance;

The specification analysis agent is used for converting the management specification item into an executable judging script, carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script, and carrying out television page compliance verification and regression evaluation.

It is to be noted that the required management specification item, DSL rule and discrimination script are directly called when the required management specification item, DSL rule and discrimination script exist in the database, then new management specification item, DSL rule and discrimination script are generated by the specification analysis agent, grammar and semantic check and security and specification check are carried out on the generated management specification item, DSL rule and discrimination script, and then the new management specification item, DSL rule and discrimination script are stored in the database. In short, the method can be directly called when a rule set with a stable version exists, if the rule is missing or a specification item is updated, an Agent2 calls a management specification retrieval module and a DSL discrimination rule management module to generate a new rule and a discrimination script, and the new rule and the discrimination script are verified and optimized to be released as a new version;

In addition, a cooperative closed loop is formed between Agent1 and Agent2 through ReAct mode. As shown in FIG. 5, when the Agent1 reasoning finds that the rule is missing or judges that the rule is abnormal, the specification item and the abnormal context are submitted to the Agent2, the Agent2 analyzes the specification item and generates or corrects a rule script, and after DSL rule management and sandbox small-scale verification, a new version is issued and fed back to the Agent1 to continue task execution. The main agent continuously collects anomalies in subsequent operations and triggers back to refine again, so that the rule system can dynamically iterate and optimize itself.

The MCP service layer comprises a plurality of functional services and provides an invokable execution interface for an intelligent agent, and comprises (1) an EPG page data acquisition service, a management mechanism and a control system, wherein the EPG page data acquisition service is used for realizing session management and remote control signal simulation of television equipment and supporting screen frame capture and snapshot generation of an EPG page, and meanwhile, the management mechanism is used for maintaining personalized acquisition flows required by televisions of different models and ensuring configurability and reproducibility of a page data acquisition process. (2) The Resource Identification Tree (RIT) service is used for carrying out structural analysis and element identification on the acquired EPG page to generate and maintain a resource identification tree, supporting tree merging, updating and consistency verification, and providing a query and retrieval interface of page elements so as to ensure the accuracy and reusability of page structure modeling. (3) The television page management specification retrieval service is used for realizing the importing of the television page management specification document, the segmentation of the specification item and the semantic vectorization and retrieval, and aligning the semantic gist of the specification item with the structural characteristics of a Resource Identification Tree (RIT) by combining with a violation case library, so as to support the automatic mapping of the specification item to the DSL rule and the discrimination script. (4) DSL discrimination rule management service, namely generating DSL rule according to management rule item, finishing grammar and semantic check of rule, combining RIT to automatically generate correspondent discrimination script, supporting version management and iterative update of rule and script and ensuring accuracy, evolution and traceability of detection rule. (5) The sandbox running environment service comprises the steps of providing an isolated execution environment, supporting compiling and sandbox environment preparation of a judging script, realizing automatic execution of the judging script based on a standard item, monitoring an execution process in real time, supporting detection reproduction of a violation interface to ensure safety, observability and reproducibility of the detection process, and (6) judging basis and report generating service, completing automatic generation and storage of the judging basis, associating RIT paths, standard items and judging script output, generating a standardized detection report on the basis, and supporting report archiving and long-term storage, so that a complete traceable link of 'item-judging script-judging result-judging basis-report' is formed.

The dual Agent layer of the embodiment of the invention consists of two agents which cooperate to form a closed loop from a standard item to detection execution, wherein a main Agent (Agent 1) is responsible for user intention analysis and detection task arrangement, scheduling page acquisition and RIT generation and update, calling published DSL rules and discrimination scripts to complete detection task execution, carrying out aggregation storage and abnormal feedback on detection results, and finally generating a compliance detection report. The intelligent agent mainly relies on the reasoning type big model to complete flow planning, task scheduling and execution control, and controllability and traceability of the detection process are ensured. The rule analysis Agent (Agent 2) is responsible for analyzing television management rules and rule entries thereof, refining key points, combining RIT to generate DSL rules and judging scripts, further executing automatic test and conflict resolution of the scripts, completing release of the rules and script versions, accumulating DSL templates, reverse modes and case data, and forming continuously optimized knowledge precipitation. The intelligent agent mainly depends on a programming large model to complete rule generation and script synthesis, and ensures the accuracy, evolution and reproducibility of a rule system. The two intelligent agents communicate through an Agent-to-Agent cooperation mechanism, a closed loop link of 'canonical item-DSL rule-discrimination script-detection execution-discrimination basis' is constructed, and full-flow intelligentization from canonical understanding to detection discrimination is realized.

4) The acquisition instruction script management specifically comprises the functions of registering, checking, searching and binding acquisition instruction scripts of different machine types/system versions, defining an acquisition flow of remote control signal simulation and page snapshot in a configurational mode, guaranteeing the configurability, the migration and the reproducibility of the acquisition process, and maintaining the alignment with a page path corresponding to a management specification/specification item for a long time through the version and compatible information maintenance.

Further, in the embodiment of the invention, the DSL discrimination rule management function module is in a key center of 'specification entry- & gt executable discrimination' in the whole television page compliance detection flow, and is under the control of (television page management specification retrieval and specification analysis agent) (RIT service, sandbox operation environment and report generation). The core responsibility of the method is to precipitate abstract specification items into declarative DSL rules (Domain-Specific Language), and after the grammar and the semantic verification are completed, the corresponding specification item discrimination script is generated by combining RIT. Finally, the rules and scripts are released in a version and traceable mode for a main agent to call, so that a closed loop link of 'management specification, specification entry, DSL rule, judging script and judging result/judging basis' is constructed, and the stability, reusability, evolution and auditability of detection logic are ensured.

In the concrete implementation, the Agent1 mainly relies on deterministic flow steps, and the MCP tool is called to realize operations such as user intention analysis, equipment session and page acquisition, RIT modeling and consistency check, sandbox discrimination script execution, discrimination basis and report generation and the like, so that the stability and traceability of a detection result are ensured, and the Agent2 internally focuses on rules and script management, and ensures the accuracy and long-term evolution of a rule system through links such as entry analysis, DSL rule generation and verification, script synthesis and test, conflict resolution and version management. In these internal flows, partial links (e.g., anomaly diagnostics, conflict resolution) embody local inference + action logic, but the whole is more biased towards the invocation and management of the engineering tool chain.

In the embodiment of the invention, a standard analysis Agent (Agent 2) is positioned as a rule analysis and optimization core in a television page compliance detection system. The method is responsible for converting management specification items issued by a broadcast television administrative department into DSL (Domain specific language) rules and discriminant scripts executable by a system, and ensuring the accuracy and stability of a rule system under different page structures and versions through automatic verification and conflict resolution. Agent2 is not directly oriented to the user, but is triggered and invoked by a main Agent (Agent 1), and a rule analysis flow is started when rule deletion, abnormality or conflict is found in the execution of a detection task. Through analysis of the specification items, generation of rules and verification and release of scripts, the Agent2 and the Agent 1 form a closed loop of 'task arrangement-rule generation-execution feedback-rule optimization', and the continuous availability, evolution and traceability of 'management specification, specification items, DSL rules, judgment of scripts, judgment results, judgment basis and detection report' of a detection link are ensured. The specific function design principle is as follows:

In the invention, the television pages with complex and various structures are converted into a hierarchical structured model by constructing a television page resource identification tree, the steps of controlled acquisition of the television pages, structured modeling of page information, generation and management of compliance discrimination rules, sandbox execution, detection report generation and the like are abstracted into services which can be called by MCP protocol, and automatic arrangement of detection tasks is realized through cooperation of an intelligent agent and a standard analysis intelligent agent, so that a closed-loop system from task configuration, page acquisition, RIT modeling, rule generation, discrimination execution to report archiving is established, and the problems that the prior art cannot adapt to supervision requirements of complex and various television pages and dynamic update are effectively solved, the full-flow automation and dynamic adaptation traceability of a compliance detection chain is improved, and the detection result can be improved.

Document Description

Method and system for detecting compliance of television page

Technical Field

The invention relates to the technical field of artificial intelligence, in particular to a method and a system for detecting television page compliance.

Background

The problem of television 'fake baby' charging is caused to be discontent for consumers and public attention. Aiming at the problems, the national broadcast television headquarter takes positive countermeasures, carries out intensive investigation and research on the problems of misleading information of consumers, poor user experience and the like on the cable television, IPTV and Internet television pages, and provides a series of rectification schemes on the basis. The realization page is concise and friendly through different standards and management specifications, so that the overall experience of a user is improved, and the transparency and convenience of service ordering and deduction flow are ensured. In a comprehensive view, the page display requirements of 3 service forms such as cable television, IPTV and Internet television can be subdivided into a plurality of major categories such as identification category, area category, function entrance category, charging policy category and the like. Each service modality has its specific requirements. The identification class requires that uniform identifications, such as a 'watch TV' identification, a 'call sign' identification, a toll free identification, etc., are presented in a significant location of the page and in a designated area. The area class requirements then specify the area ratio of the pages occupied by the different types of programs. The functional entrance class requires that a specific business area entrance is arranged on the interactive homepage, so that the user can conveniently and quickly access and use related services. The charging policy class standardizes the flow of charging and information that should be notified to the user. Although the above requirements can be manually checked or the method of template script can be adopted, the traditional manual detection and script detection have certain limitations in terms of coverage, real-time monitoring and cost effectiveness, so how to automatically and intelligently detect the television page display requirements becomes a problem to be solved.

Disclosure of Invention

The invention provides a method and a system for detecting the compliance of a television page, which are used for solving the problem that the conventional method and system can not automatically, quickly and accurately detect the compliance of the television page.

In a first aspect, the present invention provides a method of television page compliance detection, the method comprising:

creating a task according to the television page detection requirement, and determining a management specification item corresponding to the task;

collecting a television page to be detected, carrying out structural analysis and element information identification on the collected television page, and extracting page information into a hierarchical and computable data model of a page-column-element structure, namely a resource information tree RIT;

Converting the management specification item into an executable judging script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script;

Executing a judging script on each resource identification tree of the structured model under the sandbox running environment, outputting a violation result, a running log and performance indexes, and carrying out violation page reproduction according to triggering;

combining the violation results with the corresponding violation page snapshots, the resource identification tree paths and the numerical metrics, generating a judgment basis with labels and metric information, and generating a standardized report.

Optionally, creating a task according to the television page detection requirement, determining a management specification item corresponding to the task, including creating the task according to the television page detection requirement, solidifying a task context, generating a task number, wherein the task number is used for uniquely identifying the task, the task comprises concurrence, overtime, retry

strategy and authority domain, determining the management specification item corresponding to the task, acquiring the number, version, effective period and anchor point of the management specification item, and binding the acquired management specification item with the task.

Optionally, the management specification item is converted into an executable discrimination script, and grammar and semantic verification, conflict resolution, regression verification and versioning management are performed on the converted discrimination script, including:

And converting the management specification entry into an executable DSL rule by a specification analysis agent, converting the DSL rule into a discrimination script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted DSL rule and the discrimination script.

Optionally, the management specification item is converted into an executable DSL rule, the DSL rule is converted into a judging script, and the method further comprises the steps of directly calling when the required management specification item, DSL rule and judging script exist in a database, generating a new management specification item, DSL rule and judging script by the specification analysis agent when the required management specification item, DSL rule and judging script are absent in the database, and carrying out grammar and semantic verification and security and specification inspection on the generated management specification item, DSL rule and judging script and then storing the new management specification item, DSL rule and judging script in the database.

Optionally, constructing an isolated running environment for the judging script, unifying dependence, resources and random seeds, and generating a running environment number, wherein the running environment number is used for identifying a sandbox execution instance;

in a sandbox operation environment, the whole structured model locates relevant resource identification tree nodes or paths according to selectors and judgment conditions in DSL rules, performs logic operation corresponding to management specification items and threshold comparison, and outputs violation results, operation logs and performance indexes;

And (3) aggregating and sorting the detection results, organizing rule violation discrimination conclusions according to the dimensions of the page and the management specification item, marking abnormal samples and boundary conditions, and providing references for subsequent feedback and DSL rule optimization.

Optionally, the acquisition of the television page to be detected comprises the steps of establishing a controlled session with the target television equipment, and acquiring a page snapshot and a key frame.

Optionally, converting the television page to be detected into a hierarchical and computable structured data model, which comprises performing element identification and structural modeling on collected page snapshots, key frames and the like, extracting the hierarchical structure as a page-column-element, adding attributes into the extracted hierarchical structure to realize the conversion of the television page to be detected into the computable data model, wherein the data model is a resource identification tree with path and measurement information, the attributes comprise element types, display information, coordinates and area proportion, generating a resource identification tree number, performing consistency check on the resource identification tree, and correcting the resource identification tree nodes inconsistent in the hierarchical structure.

Optionally, after the standardized report is generated, the method further comprises adjusting the standard analysis agent according to false alarm, missing report and conflict conditions in the standardized report so as to improve the accuracy of matching and identifying of the standard analysis agent.

In a second aspect, the present invention provides a television page compliance detection system comprising a compliance detection master agent and a specification resolution agent, wherein,

The method comprises the steps of detecting a main intelligent agent, triggering and collecting data of a television page to be detected, extracting the television page to be detected into a hierarchical structure of page-column-element by utilizing the collected data, adding attributes into the extracted hierarchical structure, generating a resource identification tree RIT to convert the television page to be detected into a computable structural model, triggering and executing a judging script on each resource identification tree of the structural model under a sandbox running environment, outputting a violation result, a running log and a performance index, and repeating the violation page according to the triggering, combining the violation result with a corresponding violation page snapshot, an RIT path and a numerical measurement, generating a judging basis with marking and measurement information, and generating a standardized report, wherein the main intelligent agent is used for controlling the creation of the task according to the television page to be detected and determining a corresponding management specification item of the task;

The specification analysis agent is used for converting the management specification item into an executable judging script, carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script, and carrying out television page compliance verification and regression evaluation.

In a third aspect, the present invention provides a computer readable storage medium having stored thereon a computer program which when executed by a processor implements any of the methods described above.

The invention has the following beneficial effects:

The invention abstracts the screen shot of the television EPG page into a layering structure of page, column and element for the first time, introduces the attributes of element type, display information, coordinates, area proportion and the like, and constructs a computable structuring model. The method breaks through the limitation that HTML/DOM analysis cannot be used in a television environment, is directly oriented to an actual presentation interface, ensures the stability and consistency of detection input, and provides a solid foundation for rule execution and index measurement. The invention has the full-flow automation and dynamic adaptation capability driven by the agent, realizes a closed-loop system from task configuration, page acquisition, RIT modeling, rule generation, discrimination execution to report archiving through the cooperation of the main agent and the standard analysis agent, and realizes the dynamic updating and continuous optimization of the rule through the interaction of the agent in ReAct mode. Compared with the traditional mode relying on manual playback and template script, the method supports the tracing and reproduction of the cross-task and cross-time period, remarkably improves the automation degree of the detection flow and the quick adaptation capability to the standard updating, improves the interpretability and traceability of the detection result, and provides effective technical support for the double-administration anti-rebound supervision work of the television 'doll' of the broadcasting and television headquarter.

The foregoing description is only an overview of the present invention, and is intended to be implemented in accordance with the teachings of the present invention in order that the same may be more clearly understood and to make the same and other objects, features and advantages of the present invention more readily apparent.

Drawings

Various other advantages and benefits will become apparent to those of ordinary skill in the art upon reading the following detailed description of the preferred embodiments. The drawings are only for purposes of illustrating the preferred embodiments and are not to be construed as limiting the invention. Also, like reference numerals are used to designate like parts throughout the figures. In the drawings:

FIG. 1 is a flow chart of a method for detecting compliance of a television page according to an embodiment of the present invention;

FIG. 2 is a flow chart of another method for detecting compliance of a television page according to an embodiment of the present invention;

FIG. 3 is a schematic diagram of an interaction flow of a main agent according to an embodiment of the present invention;

FIG. 4 is a flowchart of main agent exception handling and rollback provided in an embodiment of the present invention;

FIG. 5 is a schematic diagram of a dual agent ReAct co-process provided by an embodiment of the present invention;

FIG. 6 is a schematic structural diagram of a television page compliance detection device based on an AI agent and a resource identification tree according to an embodiment of the invention;

FIG. 7 is a schematic diagram of a system for detecting compliance of a television page according to an embodiment of the present invention;

FIG. 8 is a schematic diagram of a resource identification tree provided by an embodiment of the present invention;

FIG. 9 is a schematic flow chart of the legal semantic parsing provided by an embodiment of the present invention;

fig. 10 is a schematic flow chart of policy DSL generation provided by an embodiment of the present invention.

Detailed Description

The present invention will be described in further detail with reference to the accompanying drawings and examples. It should be understood that the specific embodiments described herein are for purposes of illustration only and are not intended to limit the scope of the invention.

Although the existing common method for detecting the interface elements of the webpage/system, such as a scheme based on the Web crawler capturing the HTML/DOM structure, has certain convenience in the detection of the standard webpage, the method faces a plurality of adaptation problems in the real internet television EPG scene, for example, the detection task is finished mainly by relying on manual playback or semi-automatic flow, and no mechanism of agent planning, collaboration, rule generation and closed-loop execution exists, so that the method has serious lack of expansibility, automation and interpretability. And internet televisions are not typically conventional Web pages, but are native or hybrid rendering interfaces, and cannot obtain structural information through simple capture packages or DOM interfaces. The capturing process often causes inefficiency due to security (such as user key, encryption channel, etc.) related to the television system, sometimes only the interface screenshot or the picture transmitted from the bottom layer can be taken, and the complete page structure tree cannot be obtained. Moreover, the DOM abstraction does not reflect the actual visual effect of the interface, such as column salience, occlusion, resolution adaptation, etc., which makes the judgment of "visual compliance" by the DOM lack of accuracy and credibility. Moreover, the existing process is difficult to bind the information such as the boundary surface capture timestamp, the rule version, the session record and the like, the generated detection basis is mostly screenshot or structure snapshot, and a complete auditable and reproducible evidence chain is lacked. Meanwhile, the existing management system from management specification items to specific judging scripts to reports and signatures is not available, the management specification related items are difficult to carry out versioning management, and non-tamperable reports with signatures, hash values and time line tracking cannot be generated. That is, the existing page is not rendered in HTML, the capturing efficiency is low, the visual presentation structure cannot be reflected, and the session binding and audit trail capability is lacking. In order to solve the above problems, the embodiment of the invention provides a detection method based on modeling of a dual-agent architecture and a Resource identification tree (Resource IDENTIFIER TREE, RIT). RIT can convert complex page into layering and structuring model, uniformly describe navigation column, special area entry, program unit, significant mark and other elements and its measurement, and provide stable input for rule discrimination and discrimination basis generation. In the dual-Agent design, a compliance detection main Agent, namely Agent 1, is used as an execution and scheduling core and is responsible for task arrangement, page acquisition, RIT generation and updating, rule calling and detection execution and result aggregation and report generation, a specification analysis Agent, namely Agent2, is used as a rule management and optimization core, is focused on analysis and essential refining of management specifications and specification entries, generates a discrimination rule and a script based on a domain specific language (Domain Specific Language, DSL), and maintains the evolution and stability of a rule system through testing, conflict resolution and version release. Specifically, an embodiment of the present invention provides a method for detecting compliance of a television page, referring to fig. 1, the method includes:

s101, creating a task according to the detection requirement of a television page, and determining a management specification item corresponding to the task;

Before step S101, a main Agent PLANNER AGENT (hereinafter referred to as Agent 1) and a standard analysis Agent PARSER AGENT (hereinafter referred to as Agent 2) are first constructed, and through a cooperative working mechanism of the constructed main Agent and standard analysis Agent, intelligent closed-loop scheduling from user requirements, standard entry to rule generation, detection, discrimination and execution, and then to report output is implemented, so that automation, expansibility and interpretability of compliance detection of a television page are improved.

When the method is specifically implemented, a task is created according to the detection requirement of a television page, a task context is solidified, a task number job_id is generated, the task number job_id is used for uniquely identifying the task, the task comprises concurrency, overtime, retry strategies and authority domains, then a management specification item corresponding to the task is determined, the number, version, effective period and anchor point of the management specification item are obtained, and the obtained management specification item is bound with the task.

Firstly, task creation and specification item confirmation, wherein a user configures a detection object and a page range at the front end, selects specification items or a specification item set required for binding detection, and solidifies the specification items or the specification item set into a task context by Agent 1;

s102, collecting a television page to be detected, carrying out structural analysis and element information identification on the collected television page, and extracting page information into a hierarchical and computable data model of a page-column-element structure, namely a resource information tree RIT;

In specific implementation, the embodiment of the invention extracts the television page to be detected as a hierarchical structure of page, column and element, adds an attribute into the extracted hierarchical structure, and generates a unique RIT number to convert the television page to be detected into a computable structural model, wherein the structural model is a resource identification tree structure with path and measurement information, and the attribute comprises element type, coordinates, area proportion and significance;

briefly, in the embodiment S102 of the present invention, page collection and RIT modeling are performed, agent1 invokes a collection module to obtain a page snapshot and an operation log, and structural modeling and consistency verification are completed in the RIT module to generate a unique RIT number;

In specific implementation, the embodiment of the invention collects the page snapshot, the key frame and the operation log by establishing a controlled session with the target television equipment, and performs element identification and structure modeling according to the collected attributes in the page snapshot, the key frame and the operation log to generate a computable structured model. And consistency checking is carried out on the hierarchical structure of the structural model, and RIT nodes with inconsistent hierarchical structure are corrected.

S103, converting the management specification item into an executable judging script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script;

Specifically, the embodiment of the invention converts the management specification entry into the executable DSL rule through the specification analysis agent, converts the DSL rule into the discrimination script, and carries out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted DSL rule and the discrimination script.

It is to be noted that the required management specification item, DSL rule and discrimination script are directly called when the required management specification item, DSL rule and discrimination script exist in the database, then new management specification item, DSL rule and discrimination script are generated by the specification analysis agent, grammar and semantic check and security and specification check are carried out on the generated management specification item, DSL rule and discrimination script, and then the new management specification item, DSL rule and discrimination script are stored in the database. In short, the method can be directly called when a rule set with a stable version exists, if the rule is missing or a specification item is updated, an Agent2 calls a management specification retrieval module and a DSL discrimination rule management module to generate a new rule and a discrimination script, and the new rule and the discrimination script are verified and optimized to be released as a new version;

That is, in the embodiment of the present invention, various management specification entries, DSL rules and discriminant scripts may be set in advance in the database for subsequent retrieval.

S104, executing a judging script on each resource identification tree of the structured model under the sandbox running environment, outputting a violation result, a running log and performance indexes, and carrying out violation page reproduction according to triggering;

The embodiment of the invention constructs an isolated operation environment for a discrimination script, unifies dependency, resources and random seeds, generates an operation environment number env_id, wherein the operation environment number is used for identifying sandbox execution examples, under the sandbox operation environment, the whole structured model positions related RIT nodes or paths according to selectors and judgment conditions in DSL rules, performs logic operation and threshold comparison corresponding to management specification items, outputs violation results, operation logs and performance indexes, aggregates and sorts detection results, organizes violation discrimination results according to page and management specification item dimensions, marks abnormal samples and boundary conditions, and provides references for subsequent feedback and DSL rule optimization.

Namely, in the embodiment of the invention, the Agent1 calls the sandbox running environment to execute the judging script node by node on the RIT, outputs the violation results, the running logs and the performance indexes, and triggers the violation page reproduction if necessary.

S105, combining the violation results with corresponding violation page snapshots, RIT paths and numerical metrics, generating a judgment basis with labels and metric information, and generating a standardized report.

Specifically, the embodiment of the invention combines the violation results with the screenshot labels, the RIT paths and the numerical measures through the Agent1 to generate and file the standardized detection report, and feeds back false alarms, missing alarms and conflicts found in execution to the Agent2, thereby promoting the continuous optimization of a rule system.

That is, the embodiment of the invention introduces a mode of combining reasoning and actions (Reasoning AND ACTING, reAct) in the interaction of the double agents, and forms a continuous optimization closed loop between the two agents through a circulating mechanism of reasoning, action, feedback, reasoner and rerun, wherein Agent1 triggers Agent2 to generate or correct rules if the rules are missing or abnormal during the execution and detection, and Agent2 returns new version rules to Agent1 for calling after the reasoning and the rules are completed. Through the ReAct mode crossing the intelligent agent, the dynamic adaptation and continuous optimization under the conditions of management specification update and page change are realized, and a complete closed loop from management specification, specification entry, DSL rule, discrimination script, discrimination result, discrimination basis and detection report is constructed. Practice proves that the invention not only realizes the automation, reproduction and traceability of detection, but also ensures the self-evolution capability of a rule system through continuous feedback among intelligent agents, thereby providing a set of long-term stable and intelligent solution for television page compliance detection.

The method according to the embodiment of the present invention will be explained and illustrated in detail below by way of a specific example with reference to fig. 2:

The first stage is task creation and specification entry validation, where the user first configures the test object (vendor/model/region /firmware) and page scope at the front end and selects the specification entry or set of specification entries needed for binding test. Agent1 then solidifies the task context and generates a task number (job_id for uniquely identifying the one-time detection task), which contains concurrency, timeout, retry policies, and rights fields. Then, the Agent1 calls a television page management specification retrieval module to read only the number, version, effective period and anchor point of the specification item, and is strongly bound with the task to be used as a tracing base line of the detection process.

And in the second stage, page acquisition and RIT modeling, namely, after task preparation is completed, the Agent1 calls an EPG page data acquisition module, establishes a controlled session with target television equipment, simulates remote control operation according to an acquisition script, and acquires a page snapshot, a key frame and an operation log. Subsequently, agent1 sends the collection result to a Resource identification tree (Resource IDENTIFIER TREE, RIT) module, performs element identification and structure modeling on the page, generates a RIT node tree with path and metric information, and forms a unique RIT number (rit_id for identifying a page modeling result). And finally, consistency check is carried out on the RIT, incremental update and merging of the tree are carried out when necessary, and a version snapshot is generated for the key page so as to support subsequent discrimination and backtracking.

And step three, rule generation and script management, namely after page modeling is completed, agent1 firstly inquires whether a stable rule set version matched with a standard item exists in a DSL discrimination rule management module, if so, the stable rule set version is directly assembled and executed, and if the stable rule set version is missing or the standard item is updated, agent1 triggers Agent2 through interaction between agents. At this time, the Agent2 invokes the management specification retrieval module to obtain the context and case reference of the specification entry, and generates or revises a rule in the DSL discrimination rule management module, wherein the specification entry is converted into a computable decision logic by using a domain-specific language (Domain Specific Language, DSL), and further compiled into a discrimination script. The generated rules and scripts need to be checked through grammar and semantics, safety and specification, and regression verification is carried out through small-scale samples when necessary. After the above process is completed, agent2 issues the rule set as a new version and returns a rule set version number (ruleset_id@version, where ruleset_id is the rule set identifier and version is the version number) to Agent 1.

And step four, rule execution and result analysis, namely, after rule preparation is completed, the Agent1 calls a sandbox operation environment module, builds an isolated operation environment for judging the script, unifies dependence, resources and random seeds, and generates an operation environment number (env_id for identifying a sandbox execution instance). Then, agent1 applies a discriminant script to the entire RIT tree in the environment, locates relevant nodes or paths according to selectors and discriminant conditions in the rules, performs a logical operation corresponding to the canonical entry, and compares with a threshold, thereby outputting violation results, running logs, and performance indexes. Under the necessary scene, the Agent1 can also call the illegal page reproduction function of the sandbox to generate a minimum reproduction scene and a comparison material for improving the interpretability of the judgment conclusion. Finally, the Agent1 aggregates and sorts the detection results, organizes rule violation discrimination conclusion according to the dimensions of the page and the rule entries, marks abnormal samples and boundary conditions, and provides reference for subsequent feedback and rule optimization.

And step five, judging basis and report delivery, namely after detection execution is completed, the Agent1 calls a judging basis and report generation management module, combines the violation result with the page snapshot, the RIT path and the measurement value, generates a standardized judging basis list (judging basis set for identifying one-time detection), and completes signature and timestamp solidification. Then, a report_id is generated, which is used for uniquely identifying a report, and the report content covers the standard item, the rule and script version, the violation item and the discriminant basis link, and finally, the report is filed in a PDF or XLSX format and returned to the

download link. Finally, the Agent1 feeds back false alarm, missing report, conflict and RIT consistency abnormality found in the detection process to the Agent2, and the Agent2 performs rule optimization and version iteration after receiving the feedback, so as to form closed loop optimization based on ReAct modes between double agents.

In specific implementation, the main agent in the embodiment of the invention bears the core responsibility of task arrangement and execution in the television page compliance detection system, and is not only a direct carrier for user intention, but also an overall scheduling center of each functional module. The continuous evolution of the rule system is realized by receiving user configuration, driving data acquisition and modeling, calling a rule set and executing a judging script, finally generating a judging basis and a detection report, and transmitting abnormal feedback to a standard analysis agent (PARSER AGENT 2). The whole workflow takes tasks as driving, and covers five stages of task creation and specification item confirmation, page acquisition and RIT modeling, rule assembly and script management, rule execution and result analysis, judgment basis and report delivery, so that automation, reproducibility and traceability of a detection link are ensured. As shown in fig. 3, the main flow of Agent1 sequentially covers task configuration, data acquisition and modeling, rule calling, sandbox execution and report delivery, and completely presents an execution link from task input to result output.

In a specific implementation process, the main agent needs to have an exception handling and rollback mechanism. As shown in FIG. 4, when the page acquisition or RIT modeling fails, the system triggers a backoff retry, if still fails, the system marks a manual review, in the rule assembly stage, if the rule is missing, the system triggers Agent2 to generate a new rule, if the generation fails, the system rolls back to a stable version, in the sandbox execution stage, if insufficient resources or overtime abnormality are encountered, the system can adjust the resource strategy or reduce the concurrent retry, and if still fails, the system carries out manual processing. Through the series of remedy and rollback measures, the robustness and controllability of the detection flow are ensured.

In addition, a cooperative closed loop is formed between Agent1 and Agent2 through ReAct mode. As shown in FIG. 5, when the Agent1 reasoning finds that the rule is missing or judges that the rule is abnormal, the specification item and the abnormal context are submitted to the Agent2, the Agent2 analyzes the specification item and generates or corrects a rule script, and after DSL rule management and sandbox small-scale verification, a new version is issued and fed back to the Agent1 to continue task execution. The main agent continuously collects anomalies in subsequent operations and triggers back to refine again, so that the rule system can dynamically iterate and optimize itself.

The Agent2 is a rule translation and optimization core and is responsible for converting the rule entries in the management rule into rules and scripts executable by the system and ensuring the stability of the rules and scripts on different page structures and versions. The input is from Agent1, which comprises a standard item list (containing number, version, effective period and anchor point), abnormal context (false report/missing report/conflict, etc.), RIT abstract and task number, and the output is rule set version number (ruleset _id@version) and matched metadata (application range, effective time and change abstract), and the mapping link of 'standard item- & gtDSL- & gtdiscriminant script- & gtevaluation- & gtrelease' is put in storage for tracing and reproduction.

As shown in fig. 6, in the core flow, agent2 first performs structural analysis on the specification entry, refines the decision gist (element category, selector path, metric index, threshold/tolerance, etc.), generates a DSL (domain specific language) rule draft from this, and compiles the rule draft into a decision script draft. Three thresholds, ① grammar and semantic checks (field integrity, selector resolvable, threshold range aligned with RIT index), ② security and specification checks (white/black list, resource and dependency limits, network forbidden policies), ③ small scale regression evaluations (statistical coverage/false positive rate/false negative rate on sandboxed sampling RIT set) were then performed in order. Any threshold is not met, i.e., a return revision, which may involve threshold trimming, selector adjustment, conflict resolution (context entries, repetition rules, override relationships), and rule splitting/merging, and requests Agent1 to supplement the context if necessary, until it is met. After the evaluation reaches the standard, agent2 completes release and warehousing of the version, namely ruleset _id@version is generated, product hash is calculated and metadata is solidified, and meanwhile, links of the items, DSL, script, evaluation and release are written into a pedigree library to support subsequent discriminant tracing and historical reproduction. And meanwhile, the Agent2 precipitates effective experiences (DSL templates, reverse modes and case-threshold mapping) into a knowledge base, improves the efficiency and stability of subsequent rule generation and verification, and forms a long-term evolution mechanism of 'remixing-reissuing'.

The invention provides a television page compliance detection system based on an AI intelligent agent and a resource identification tree. The system mainly detects television page compliance through double intelligent agents, wherein the main intelligent agent is used for creating a task according to a television page to be detected and determining a management specification item corresponding to the task, converting the management specification item into an executable judging script, carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script, extracting the television page to be detected into a layering structure of page-column-element, adding an attribute into the extracted layering structure, generating a unique RIT number, converting the television page to be detected into a computable structuring model, wherein the structuring model is a resource identification tree with path and measurement information, the attribute comprises element types, coordinates, area proportion and significance, combining a rule analysis intelligent agent violation result with corresponding rule page snapshot, RIT path and value measurement,

generating a judging basis with label and measurement information, and generating a standardization report, and using the analysis intelligent agent to trigger the rule analysis intelligent agent to execute the rule analysis result under the environment of each rule page under the condition of the rule analysis rule, and carrying out statistics, and triggering the rule analysis script according to the performance of each rule.

In specific implementation, as shown in fig. 7, the detection system of the invention is divided into a data base layer, an MCP service layer, a dual agent layer and a visualization layer from bottom to top, and the compliance detection of the television page is realized through the cooperation of data support, tool service, agent reasoning cooperation and a user interaction interface. Wherein, the

The data base layer is used for centrally storing core data in the running process of the system and comprises EPG page images, resource Identification Tree (RIT) structural results, television management standards and standard entries, DSL rules and corresponding judging scripts and results and judging bases of conventional detection tasks. The layer provides unified data support, comparison and traceability for compliance detection, and ensures traceability and reproducibility of detection results.

The MCP service layer comprises a plurality of functional services and provides an invokable execution interface for an intelligent agent, and comprises (1) an EPG page data acquisition service, a management mechanism and a control system, wherein the EPG page data acquisition service is used for realizing session management and remote control signal simulation of television equipment and supporting screen frame capture and snapshot generation of an EPG page, and meanwhile, the management mechanism is used for maintaining personalized acquisition flows required by televisions of different models and ensuring configurability and reproducibility of a page data acquisition process. (2) The Resource Identification Tree (RIT) service is used for carrying out structural analysis and element identification on the acquired EPG page to generate and maintain a resource identification tree, supporting tree merging, updating and consistency verification, and providing a query and retrieval interface of page elements so as to ensure the accuracy and reusability of page structure modeling. (3) The television page management specification retrieval service is used for realizing the importing of the television page management specification document, the segmentation of the specification item and the semantic vectorization and retrieval, and aligning the semantic gist of the specification item with the structural characteristics of a Resource Identification Tree (RIT) by combining with a violation case library, so as to support the automatic mapping of the specification item to the DSL rule and the discrimination script. (4) DSL discrimination rule management service, namely generating DSL rule according to management rule item, finishing grammar and semantic check of rule, combining RIT to automatically generate correspondent discrimination script, supporting version management and iterative update of rule and script and ensuring accuracy, evolution and traceability of detection rule. (5) The sandbox running environment service comprises the steps of providing an isolated execution environment, supporting compiling and sandbox environment preparation of a judging script, realizing automatic execution of the judging script based on a standard item, monitoring an execution process in real time, supporting detection reproduction of a violation interface to ensure safety, observability and reproducibility of the detection process, and (6) judging basis and report generating service, completing automatic generation and storage of the judging basis, associating RIT paths, standard items and judging script output, generating a standardized detection report on the basis, and supporting report archiving and long-term storage, so that a complete traceable link of 'item-judging script-judging result-judging basis-report' is formed.

The dual Agent layer of the embodiment of the invention consists of two agents which cooperate to form a closed loop from a standard item to detection execution, wherein a main Agent (Agent 1) is responsible for user intention analysis and detection task arrangement, scheduling page acquisition and RIT generation and update, calling published DSL rules and discrimination scripts to complete detection task execution, carrying out aggregation storage and abnormal feedback on detection results, and finally generating a compliance detection report. The intelligent agent mainly relies on the reasoning type big model to complete flow planning, task scheduling and execution control, and controllability and traceability of the detection process are ensured. The rule analysis Agent (Agent 2) is responsible for analyzing television management rules and rule entries thereof, refining key points, combining RIT to generate DSL rules and judging scripts, further executing automatic test and conflict resolution of the scripts, completing release of the rules and script versions, accumulating DSL templates, reverse modes and case data, and forming continuously optimized knowledge precipitation. The intelligent agent mainly depends on a programming large model to complete rule generation and script synthesis, and ensures the accuracy, evolution and reproducibility of a rule system. The two intelligent agents communicate through an Agent-to-Agent cooperation mechanism, a closed loop link of 'canonical item-DSL rule-discrimination script-detection execution-discrimination basis' is constructed, and full-flow intelligentization from canonical understanding to detection discrimination is realized.

The visual layer of the embodiment of the invention provides a unified man-machine interaction entrance for an end user and bears the whole flow operation and observation from the detection task creation to the result presentation. The capability of the dual-agent layer and the MCP service layer is output to the outside in an operable mode through task configuration, process monitoring and result presentation, and the transparency, traceability and reproducibility of television page compliance detection are ensured. The system comprises a user portal for logging in and identity verification and completing creation and configuration display of television compliance detection tasks (such as target equipment/page range, associated specification items, DSL rules and distinguishing script versions, and the like), a detection process control providing detection control console and real-time monitoring capability, supporting task starting, suspending, stopping and retrying, showing running states, key logs and sandbox playback trigger portals, a television specification management centralizing

presenting management specification and specification items, supporting retrieval, comparison and version viewing, associated displaying DSL rules and distinguishing scripts derived from the specification items, a detection result and a basis presenting intuitively displaying judging basis and reporting downloading in the forms of rule violation item list, page snapshot labeling, RIT paths, sandbox playback links, and the like, and a system management realizing unified configuration of user and role rights, task and resource quota, auditing and retention strategy and rule/script version visibility, and guaranteeing system stability and compliance operation.

Further, in the embodiment of the present invention, the MCP service layer specifically includes:

The EPG page data acquisition function module is positioned at an entrance position of the full-flow compliance detection of the television page, is responsible for executing an interaction path according to an acquisition instruction script at a real television terminal, acquiring page snapshots/key frames and recording an operation process to form reproducible and traceable original page data, and is a basic data source for the subsequent construction of a Resource Identification Tree (RIT), the execution of DSL rules and discrimination scripts and the reproduction of an illegal interface. The module initiates remote control signals to the set top box, the intelligent television and the simulation terminal in an automatic and configurable mode and completes page grabbing, so that the problems of high cost and inconsistency caused by manual operation and manual screenshot are avoided. The acquisition output uniformly contains metadata such as task numbers, session numbers, time stamps, arrival paths, interaction logs and the like, is convenient to be aligned with RIT nodes, standard entries and discrimination script output, and is finally used for forming a traceable discrimination basis. The actual collection flow is usually arranged according to the page paths related to the specification items (such as a "first page head screen", "pay Fei Shanbian special area", "order entry", etc.), so as to ensure that the page elements corresponding to the management specification requirements are completely recorded. The core function is realized by the following tool functions:

1) The function of the session management of the television equipment is to establish a controlled session with a designated television terminal, complete basic state detection and context binding and provide a stable channel for subsequent remote control operation and page acquisition. Typical inputs are television equipment information (model, resolution, region, system version), task numbers, tenant identifications, acquisition script numbers (corresponding scripts can be assembled by agents in a session preparation stage), typical outputs are session numbers, equipment states (online/offline, current resolution, network delay), session context metadata (task/session binding, equipment fingerprints, log channels), explanation that the session metadata need to be put in storage and bound with the task numbers, guarantee consistency with subsequent RIT and discrimination basis backtracking, and health detection/heartbeat interfaces are provided, so that agents 1 can execute overtime, reconnection and recovery conveniently.

2) The remote control signal simulation function is to simulate the selection/sliding of the keys and coordinates of a remote controller and drive the navigation and interaction of the EPG page, so as to form a reproducible page arrival path. Typical inputs are session number, keystroke sequence (e.g. "home→arrow key→confirm", supporting keystroke interval and retry strategy), touch event (coordinate click/slide, throttle and debounce configuration), typical outputs are operation log (timestamp, step number, success/failure, number of retries and failure reason), arrival page identity (e.g. "head of page screen", "pay Fei Shanbian area", "order entry", etc.), explaining that critical jumps directly related to canonical entries should automatically trigger page snapshots, ensuring detection recheck, path traceable, providing power etc. retries and recovery strategy back to anchor page (e.g. "HOME) upon failure.

3) The screen frame capturing and page snapshot functions are to collect a screenshot or key frame sequence of a current page to form original page data for RIT construction and rule discrimination. Typical inputs are session number, acquisition mode (screenshot/keyframe sequence), acquisition duration, frame rate (effective when keyframe sequence), region of interest (ROI clipping for key region capture of high resolution device), typical outputs are acquisition file path (picture or keyframe sequence storage path/URI), media metadata (resolution, color space, file checksum), timeline metadata (timestamp, trigger reason, associated with oplog step number), illustrate that acquisition file needs to be strongly bound with task number/session number, and record source path (provided by "remote control signal simulated" arrival page identification and oplog) to support the generation of the RITs node location and discrimination basis, output of the re-fingerprint (hash value) for facilitating buffer multiplexing and repeated scene identification.

4) The acquisition instruction script management specifically comprises the functions of registering, checking, searching and binding acquisition instruction scripts of different machine types/system versions, defining an acquisition flow of remote control signal simulation and page snapshot in a configurational mode, guaranteeing the configurability, the migration and the reproducibility of the acquisition process, and maintaining the alignment with a page path corresponding to a management specification/specification item for a long time through the version and compatible information maintenance.

Typical inputs are ① script registration subfunction, script name, equipment matching rule (model/resolution/region), step definition (key sequence and expected page anchor), script version, file check value, ② script check subfunction, instruction script number, idle running session number (idle running check in controlled session)

③ The script searching/binding subfunction comprises a television equipment information or collecting script number and task number, typical output comprises ① registering/updating results, namely collecting script number, script version and verification state, ② verification results, namely verification report (pass/fail, fail step and recovery proposal), ③ searching/binding results, namely matched script number and binding metadata (task-script binding relation), and the instruction collecting script is only used for defining and managing a collecting flow, the execution of the instruction collecting script is still completed in three functions of 'television equipment session management + remote control signal simulation + screen frame capture and page snapshot', and script versioning and compatibility matrix (model/resolution /system version) changing information is recorded.

The Resource Identification Tree (RIT) functional module in the embodiment of the invention is a page structure modeling and retrieval engine in television page compliance detection. The method has the responsibility of converting the acquired EPG page snapshot and metadata into a tree structure model and providing the capabilities of tree management, structure consistency check, page element inquiry, version management and the like around the model. The RIT provides unified, reusable and retrievable structured input for subsequent DSL rules and discrimination scripts, and provides semantic annotation and positioning interfaces for page elements (such as navigation column columns, area entries, VIP/free identifiers and the like) related to the standard entries, so that accurate reference can be conveniently carried out in detection execution and discrimination basis generation. The RIT does not generate a compliance/non-compliance conclusion or is not responsible for flow arrangement, the compliance judgment is completed in a sandbox environment by a judging script, and the RIT only provides structural data and retrieval support.

The EPG page recognition function of the embodiment of the invention is to recognize the page type and extract the UI elements and text elements in the page, thereby providing element candidates for subsequent structural modeling.

Typical inputs are page snapshot path (picture/key frame sequence), acquisition metadata (task number, session number, timestamp, television equipment information), recognition model version (OCR/detection model version number);

Typical outputs are page type (e.g., home page head screen, application center, order information page, payment page, etc.), candidate RIT node set, i.e., made up of several "candidate RIT nodes", each containing attribute information including element type (navigation item /section entry/recommendation bit/program unit/icon/text label, etc.), normalized coordinates and size, OCR text/semantic label (e.g., "free" "VIP" "payment monolithic"), visibility (optional), confidence, source metadata (timestamp, frame number, model version, session number);

The candidate RIT nodes are preliminary identification result items produced in the identification stage, are not subjected to screening, de-duplication, merging, hierarchical organization and consistency check of 'television EPG page structure modeling', are different from the final RIT nodes, and may contain repetition, false detection or incomplete information. The candidate set of RIT nodes may be incorporated into the RIT node tree after screening/de-duplication/merging/complementing/hierarchical organization/consistency checking. The judging script only depends on the RIT node tree to complete the judgment of the standard item, and forms a judging basis according to the judgment.

In the modeling of the television EPG page structure in the embodiment of the invention, the function is to organize a candidate RIT node set into a RIT node tree according to the space and hierarchy relation, so as to form a unified structural representation.

Typical inputs are candidate set of RIT nodes, page metadata (task number, session number, resolution, arrival path).

Typical outputs are RIT number, complete RIT node tree (including node ID, parent-child relationship/hierarchy, element type, normalized coordinates and size, text/label, visibility, confidence, RIT path, etc.), modeling summary (total number of nodes, tree depth, key semantic anchor);

the method is characterized in that semantic node labeling is carried out on common elements (navigation bar, special area entrance, recommendation position, program unit, VIP/free mark and the like), so that the searching is facilitated, and a compliance conclusion is not output in the step.

The RIT tree management function of the embodiment of the invention is to add, delete, merge and update the existing RIT, and maintain the binding relation between the RIT, the acquisition product and the semantic annotation.

Typical inputs are RIT number, operation type (add node/update node/delete node/merge update), node change list (node ID, attribute change: coordinates, labels, visibility, text, etc.), associated metadata (task number, session number, page snapshot path);

typical outputs are operation results (success/failure, failure reasons), updated RIT numbers or change sequence numbers, change abstracts (affected node ID list, anchor change description);

the method is characterized in that the merging update is used for unifying the same page under different acquisition time periods or slight resolution differences, and only the structure and the labeling are affected.

The RIT consistency check function of the embodiment of the invention is to check the structural integrity and constraint consistency of the RIT and find out the relation between the model gap and unreasonable relationship.

Typical inputs are RIT number, check configuration (check item switch: coordinate range, overlap conflict, hierarchical rationality, necessary anchor presence, etc.);

Typical outputs are consistency check results (pass/fail), issue list (issue type, involved node ID, RIT path, severity level, suggested repair actions), check log (time consuming, rule version);

The method and the device have the advantages that consistency check faces to the structure and the constraint, and the judgment script is recommended to be repaired and executed when the consistency check fails, so that the reliability of the judgment basis is prevented from being influenced.

The page element inquiry and retrieval function of the embodiment of the invention is to provide element level retrieval capability based on RIT, and support the judging script to locate and read node attributes according to the requirement of standard items.

Typical inputs are RIT number, search conditions (combinable), element type (navigation item/section entry/recommended bit/program unit /icon/text label, etc.), semantic label ("free" "VIP" "pay-per-view" ", etc.), text keyword (OCR text inclusion/matching), location range (normalized coordinates or area threshold), hierarchical path (RIT path expression), visibility (visible/translucent/occlusion);

typical outputs are hit node list (node ID, RIT path, element type, coordinates and size, text/label, visibility), hit statistics (number of hits, number of de-duplicated);

The method includes the steps that a function does not directly give a compliance conclusion, a judging script calls the function in a sandbox to obtain a node set, then the judgment of a specification item is completed by combining with DSL rules, and RIT paths and attributes are quoted in a report to form a judging basis.

The version management function of the embodiment of the invention is to carry out versioning management on RIT, support version creation, inquiry and rollback, and ensure traceability and reproducibility of task level and item level.

Typical inputs are RIT number, type of operation (create version/get version/rollback to version/list history), optional parameters (target version number, version remark);

typical outputs are version operation results (success/failure, failure reasons), version numbers or version lists, RIT references (snapshot identification) for specified versions;

the method and the system have the advantages that version management is used for time point retention and backtracking, and difference comparison and compliance judgment are not provided in the module, so that out-of-range is avoided.

In the embodiment of the invention, the television page management specification retrieval function module serves as a knowledge base and a bridge in the whole detection flow, is responsible for importing and dividing a management specification issued by a broadcasting television administration department into a quotable specification item, solidifying metadata such as a number, a version, a validation period, an anchor point and the like for each item, completing semantic vectorization and retrieval/comparison, and simultaneously maintaining a rule-breaking type case library associated with the item as a rule generation and verification reference; in operation, the specification analysis agent initiates semantic search/comparison to the module based on task description or page phenomenon to locate relevant specification items, synthesizes DSL rules and discriminant scripts according to the semantic search/comparison, and in a rule recycling stage (namely, after the first edition rule/script generation and execution by sandboxes or small scale, diagnosis, threshold and condition correction, necessary rule splitting or merging are carried out for false alarm/missing report and conflict based on abnormal feedback, RIT consistency information and case library of the main agent, and regression evaluation is completed in the sandboxes, release of version is entered when reaching standards, iteration is continued when reaching standards, effective experience is deposited into DSL templates, anti-mode and traceable links mapped by the items), the main agent reads only the reference item numbers, version and effective period in the task configuration and reporting stage, binds detection tasks and items to support report traceable sources, and provides the authority of item sources and traceable references (without participating in page structure modeling or fitting conclusion

generation) to RIT service and sandboxes operation environment for downstream, and output text vectors and metadata thereof are associated with the traceable references and case sources, the judged script is quoted when executing and generating the judging basis, so that the closed loop of 'management specification- & gt specification item- & gt DSL rule- & gt judging script- & gt judging basis' is penetrated, and the authority of item sources, accurate semantic matching, traceable quote and reproducible history are ensured. The specific functions are realized by the following tool functions:

The function of importing the management specification document in the embodiment of the invention is to import the management specification document issued by the broadcasting and television authorities, solidify sources and metadata, and form an authoritative specification source. Typical inputs are document file path or upload number (e.g., portable document format, hypertext markup document, word processing document, etc.), document metadata (issuing agency, document number, version number, validation period, applicable region, issue/revision date), canonical document number, metadata record (version number, validation period, applicable region, issuing agency check status), import check result (success/failure, failure reasons such as missing pages, failing to parse fonts, etc.), source and integrity information (source link digest, file integrity hash value/electronic signature check value), page number/chapter coarse granularity anchor index (for reference by subsequent entry anchor points), explaining that only document layer import and metadata solidification are completed, no entry segmentation or semantic processing is performed, source trace in reporting and discriminating basis.

The function of the specification entry segmentation of the embodiment of the invention is to analyze the imported management specification document into the quoteable specification entries, and generate a unique number and an accurate anchor point for each entry. Typical inputs include a canonical document number, a segmentation strategy (title hierarchy, numbering rules, layout features and regular matching rules), typical outputs include a canonical item list (canonical item number, item text, application scope/object, belonging chapter /page number/paragraph anchor, version number, validity period, status), a segmentation report (segmentation success rate, items requiring manual review, suspected page-crossing item prompts), and an explanation that the canonical item number is unique and stable in the same version, and an anchor point is used for visual positioning and audit tracing.

The function of the canonical item vector generation and comparison of the embodiment of the invention is to carry out semantic vectorization on canonical items, provide semantic retrieval and item comparison, support canonical analysis agent to accurately position items and develop rule synthesis. Typical inputs are vector generation (canonical item number list), semantic retrieval/comparison, query text (such as navigation column number is less than or equal to 15 and free area of special area of first page is more than or equal to 50 percent) or item pairs to be compared, selectable filtering conditions such as version number, effective period, applicable region and issuing mechanism, typical outputs are vector generation results (canonical item number, semantic vector, vector model version and generation time), semantic retrieval results, candidate canonical item list (canonical item number, item text abstract, relevance score, affiliated version /effective period and anchor point), item comparison results (similarity score and difference key abstract), RIT alignment prompt (page element category and measurement dimension commonly associated with the item). The method comprises the steps of only outputting semantic relativity of an item layer, not generating rules/scripts and not outputting compliance conclusions, and completely recording a link from a search condition to an item hit for repeated disc at a rule remixing stage.

The function of the rule-breaking type case library of the embodiment of the invention is to maintain typical rule-breaking types and example samples bound with the rule entries, and provide references for rule generation and verification of the rule remixing stage. Typical inputs include rule-breaking type labels (such as "charge package source exceeding", "lack of free entry", "VIP identification insignificant"), associated canonical item numbers, example resources including page snapshot paths, associated resource identification tree nodes/path references, explanatory text, sample attributes including positive/negative/boundary samples, acquisition time, television equipment information, source task numbers, typical outputs including case numbers, index labels (rule-breaking type, canonical item numbers, version /validity period), sensitive desensitization marks, case search results, i.e., return of case list (snapshot summary, resource identification tree path excerpt, remark) by item/label/keyword, explanatory case libraries are used for reference and comparison, not directly used as discrimination basis for single detection, their binding with item/version/validity period guarantees misinformation and traceability of knowledge references, and provide error/missing report clusters and regression evaluation samples for rule-back stage.

Further, in the embodiment of the invention, the DSL discrimination rule management function module is in a key center of 'specification entry- & gt executable discrimination' in the whole television page compliance detection flow, and is under the control of (television page management specification retrieval and specification analysis agent) (RIT service, sandbox operation environment and report generation). The core responsibility of the method is to precipitate abstract specification items into declarative DSL rules (Domain-Specific Language), and after the grammar and the semantic verification are completed, the corresponding specification item discrimination script is generated by combining RIT. Finally, the rules and scripts are released in a version and traceable mode for a main agent to call, so that a closed loop link of 'management specification, specification entry, DSL rule, judging script and judging result/judging basis' is constructed, and the stability, reusability, evolution and auditability of detection logic are ensured.

The DSL of the embodiment of the invention ensures that the requirement of the compliance entry can be clearly expressed in a mode of declarative constraint, structural selector and multiplexing operator, ensures that the rule is strongly coupled with the RIT page structure and

supports version governance. Compared with the method for judging the compliance based on the page image by directly relying on the large model, the method has the remarkable advantages of 1) the interpretability, the rule, the item and the judging basis are clear in relation and can be traced and verified, 2) the stability, the detection logic is independent of the training drift of the large model, the long-term stability of an execution result is ensured, 3) the consistency is realized, the rule can keep uniform execution semantics on different equipment and samples, 4) the engineering is facilitated, the gray level release, the version rollback and the regression evaluation are facilitated, and the treatment capability is improved. The generated discrimination script is a bridge for actually detecting the landing of DSL, and is responsible for interacting with RIT, calling a calculation operator, processing boundary conditions and collecting evidence, ensuring that the rule can be executed and reproduced in the sandbox environment and outputting discrimination basis (such as RIT path, page snapshot mark, index calculation value and the like) capable of being directly referenced.

In the implementation process, the programming type large model is mainly used for two links, namely providing template filling and parameterization support when DSL rule draft is generated and assisting in generating template codes when judging script synthesis. All large model generated products must pass DSL checking (grammar/semantic) and script security verification before entering the version release process. Through the mechanism of 'model assistance + engineering quality entrance guard', the DSL discrimination rule management function module can ensure the generation efficiency and the accuracy, evolution and reproducibility of the rule system. The specific functions are realized by the following tool functions:

The DSL rule generation function of the embodiment of the invention is to generate an interpretable DSL rule draft according to the television management specification item and the page type. Typical inputs are canonical entry number, entry body, page type (e.g., first page header, navigation bar, payment page), generation parameters (threshold settings, tolerance policies, whether large model draft generation is invoked), association case number (optional). Typical outputs are DSL rule text (containing rule name, applicable page, selector condition, decision index and threshold, specification information), entry to DSL mapping record (entry number, entry version, validation period to DSL rule number correspondence), report generation (pattern generation, manual validation or not, risk potential prompt). The method comprises the steps of generating a rule draft only, compiling and executing only after entering a subsequent verification link, and supporting quick instantiation based on a template or calling a programming large model to carry out rule draft.

The function of DSL grammar checking in the embodiment of the invention is to check the grammar level of DSL rule text, and ensure that the format, keywords and structure are consistent with the Schema definition. Typical inputs are DSL rule text, DSL SCHEMA version. Typical outputs are inspection results (pass/fail), problem list (lack of fields, redundancy selector, undefined keywords, etc.), repair advice (autocompletion, example modification scheme). The grammar checking is a necessary link before DSL enters compiling, and a judging script cannot be generated when DSL does not pass through the link.

The DSL semantic verification function of the embodiment of the invention is to verify the consistency of DSL rules with RIT structures and specification items at the semantic level, so as to ensure that a selector can be solved, an index can be calculated and a threshold value is reasonable. Typical inputs are DSL rule text, RIT structure summary (available node type, label, attribute, metric), specification entry metadata (number, version, validation period). Typical outputs are check results (pass/fail), problem list (selector cannot match, index does not support, threshold out of range, etc.), repair guidelines. After passing the semantic verification, DSL rules can enter the script generation stage, and the DSL rules are not modified by the generation link needing to be returned.

The function of generating the judging script based on DSL and RIT in the embodiment of the invention is to compile the DSL rule passing the verification into the standard entry judging script, so that the judging script can call RIT in the sandbox environment to calculate and judge. Typical inputs are DSL rule text, target operating environment (e.g., python), script generation options (whether to generate unit test, whether to enable evidence collection interface). And (3) judging script source codes (including RIT calls, index calculation, judging logic and evidence collection logic), item and source code line number mapping (tracing), and script dependency lists (operator library and interface version). The generated script must output the judging conclusion and the judging basis required intermediate quantity (RIT path, snapshot mark and calculated value) simultaneously, so as to ensure that the result can be reproduced and tracked.

The script security and standardization checking function of the embodiment of the invention is to scan the judging script safely and standardly, so as to avoid the script calling unauthorized interfaces or executing unauthorized operations. And (3) typically inputting judging script source codes, security policies (white list/black list) and sandbox environment constraints. Typical outputs are inspection results (pass/fail), list of violations (rule call points, description), repair advice. The security check is a necessary link before the script is issued, and the failed script cannot be archived and issued.

The rule issuing function of the embodiment of the invention is to package DSL and discriminant script passing through the verification into a rule set and issue the rule set to a version library for the main agent to call. Typical inputs are DSL rule text, discriminant script source code, specification entry number and metadata (version, validation period), applicable page range, equipment range, validation time, release

notes. Typical outputs are rule set number, version number, release record (containing metadata, scope of application, hash signature), release result (success/failure, failure reason). The method has the advantages that the issued rule set is the only authoritative version, the downstream task can only refer to the issued version, and the stability of detection tracing is guaranteed.

The rule version management function of the embodiment of the invention is to carry out version maintenance on published rules, including version reading, version comparison and rollback. Typical inputs are rule set number, target version number (read or rollback), comparison version number (difference comparison), description of the operation. Typical outputs are reading DSL text, discriminating script source code, metadata (clause map, applicability, signature). The comparison version difference summary (new/modified/deleted/threshold changed), potential impact report. The new active version number and scope of influence description (affected task/entry/page) are rolled back. The version management ensures that the evolution and rollback of the rule system can form a complete rule lineage link.

In specific implementation, the sandbox operating environment functional module plays a role of an execution and verification center in the whole television page compliance detection flow, and receives the judging script generated by the DSL judging rule management functional module and completes compiling, executing, monitoring and reproduction in a controlled environment. The primary function of the module is to construct an isolated and reproducible running environment for judging the script, uniformly restrict execution dependence, computing resources and security policies, and ensure that the running results of the script are consistent in batches of different time, equipment and tasks. At the execution level, the module supports running a discrimination script on single or batch RIT page trees, and automatically outputs a discrimination result and a corresponding discrimination basis (such as a rule-breaking item list, a RIT path, a screenshot mark and the like). Through a process monitoring mechanism, the module can output logs, indexes and state events in real time, and is convenient for abnormality positioning and SLA guarantee. When the suspected violations are detected, the sandbox operating environment can trigger the violating interface to reproduce, replay the key interaction sequence and generate a 'minimum reproduction scene', so that the transparency and the interpretability of the judging process are ensured. From the whole system, the sandbox operation environment function module is an execution bridge for connecting the specification item, DSL rule, judging script and detection result, judging basis and compliance report. Compared with the method that a conclusion is directly given by a large model, the sandbox execution ensures the certainty and traceability of regular operation, the same input must obtain the same output, and the operation process and the result can be reproduced and audited, so that a compliance detection closed loop is formed. The specific functions are realized by the following tool functions:

The function of the embodiment of the invention for compiling the discrimination script and preparing the sandbox running environment is to construct an isolation running environment for a specified rule set, unify dependency locking, random seeds and resource allowance, precompiled the discrimination script, and ensure that the cross-task execution result is consistent with the cross-equipment execution result. Typical inputs are rule set number, version number, runtime configuration (Python version, dependency package list, CPU/memory allowance, timeout threshold, random seed). Typical outputs are running environment number, environment representation (dependency list, resource limitation, random seed), precompiled product address, environment preparation log. The method comprises the steps of unifying a 'preparation-running' base, ensuring the consistency of tasks of different batches, and providing reproducible logs and environment hash fingerprints when the tasks fail.

The function of the automatic execution of the television management specification item judging script is to run the judging script on the appointed RIT page tree, complete the automatic detection corresponding to the specification item and output a violation list and a judging basis. Rule set number, version number, RIT number, execution allowance (CPU/memory/node number/execution time). Typical outputs are violation results (rule number, node number, violation cause, bounding box/text/index value), execution statistics (time consuming, number of scanned nodes, memory usage), execution log. The method is characterized in that a judging script directly acts on RIT structures and attributes (such as column numbers, area of areas and mark significance), and a detection conclusion is aligned with a report generating service to ensure reproducibility and auditability.

The function of the judging process monitoring of the embodiment of the invention is to output logs, states and performance indexes in real time in the script running process, and support the diagnosis of the executing process and the monitoring of the service quality. Typically, a task number or an execution result number. Typical outputs are a stream log (timestamp/level/content), progress event (completed/total /rate), running index (CPU, memory, number of rule hits, timeout times), exception context fragments. The process monitoring provides basis for task abnormality positioning and operation and maintenance guarantee, and can be integrated with a platform monitoring and alarming system.

The function of detecting and reproducing the illegal interface in the embodiment of the invention is to replay key interaction operation around the detected illegal point and generate the minimum reproduction scene, and output a comparison report and a necessary evidence packet. Typical inputs are violation result list (containing RIT number, node number, selector information), reproduction plan (interaction step, playback range, termination condition). Typical outputs are a recurring report (key frame control, differential labeling, threshold and index details), a minimum recurring scene package (RIT slice, screenshot and interaction sequence necessary), and a diagnostic specification. The method and the device ensure that the detection conclusion can be rechecked, and are typically applied to the

explanatory scenes of entry detection such as ' the free special area of the first page is more than or equal to 50% ' the number of navigation column columns is less than or equal to 15 '.

In specific implementation, in the embodiment of the invention, the judging basis and report generation management function module serves as a terminal hub of 'result solidification and external delivery' in the television page compliance detection flow. The method has the duty of structurally processing the judging result and the process evidence output by the sandbox running environment into two products which can be verified, traced and submitted, namely, judging basis (including page snapshot cutting and marking, RIT path and node interception, metric value snapshot, execution log abstract and the like), and standardized detection report (PDF/XLSX facing supervision/manufacturer /internal examination). The module takes the entry number, DSL rule and script version and RIT path as main indexes, uniformly completes the generation and storage of evidence and report, and carries out signature, hash and time stamp solidification on all products, thereby ensuring that the same input can be reproduced and audited at any time point. The module is connected with a detection link of 'normal item-DSL rule-discrimination script-sandbox execution', and provides detection document materials which can be delivered outwards by authority numbers and signature fingerprints downwards, so that tight alignment of conclusions and fact materials is ensured, and stable basis is provided for follow-up rechecking and archiving, thereby closing a full chain of 'management rule- & gt normal item- & gt DSL rule- & gt discrimination script- & gt discrimination result- & gt discrimination basis- & gt detection report'. The specific functions are realized by the following tool functions:

The function of the generation of the judging basis of the embodiment of the invention is to process the sandbox execution result into the judging basis, which comprises page screenshot cutting and marking, RIT path and node slicing, numerical measurement snapshot and running log abstract. Typical inputs are execution result number, violation result list (entry number, rule number, RIT number/node, bounding box or polygon, index and threshold), original page snapshot path, RIT data reference, rendering parameters (labeling style, clipping margin, output format). Typical outputs are discriminating according to the manifest (screenshot cut file path, annotation file path, RIT slice file path, numerical metrics file, running log fragment), according to the index (entry number, rule number, script version, RIT number/node, timestamp). The method ensures that 'everywhere is not compliant and why is not compliant' is readable at the image and structure level and corresponds to the specification items one by one.

The function of the storage of the judging basis in the embodiment of the invention is to store the generated judging basis, and solidify the signature and the time stamp to form a traceable basis list. Typical inputs are discriminating by manifest path, by index (entry number, rule number, script version, RIT number/node, execution time), curing algorithm (signature and hash type). Typical outputs are by manifest numbering, curing results (signature, hash digest, timestamp), warehousing returns (storage location, retention policy). The method has the advantages that the discrimination basis is strongly bound with the specification item, DSL rule and discrimination script version, and long-term consistency and traceability are guaranteed.

The function of the detection report generation of the embodiment of the invention is to assemble the standard entry quotation, rule and script version, sandbox execution conclusion and discrimination basis list into a standardized detection report. Typical inputs are task number, rule-breaking result summary (by item, page aggregation), identification by list number, report template type (supervision /manufacturer/internal examination), report parameters (cover information, header footer style). Typical outputs are report file path (PDF, XLSX), report directory and content index (entry number and version/validation period, rules and script version, RIT path and node, per hash digest). The report text automatically displays the numerical measurement of key entries (such as 'free special area is more than or equal to 50 percent', 'navigation column number is less than or equal to 15') and the judgment logic abstract thereof, and the base fingerprint is embedded in the header footer, so that the external verification is facilitated.

The detection report storage function of the embodiment of the invention is to carry out warehousing, signature and hash curing on the generated detection report, and establish index binding with task numbers, items, rules and discrimination basis to support long-term archiving. Typical inputs are report file path (PDF/XLSX), associated metadata (task number, entry number and version/validation period, rules and script version, discriminant basis list number), curing algorithm. Typical outputs are report archive number, curing results (signature, hash digest, timestamp), warehouse-in receipt (storage location, retention period, access rights configuration). The report and the judgment are based on the bidirectional index, and the full link tracing and cycle-crossing rechecking of the conclusion-basis-execution result are ensured.

The MCP service layer tool function design is specifically shown in table 1.

TABLE 1 MCP service layer tool function design Table

[Image Omitted]

[Image Omitted]

Specifically, the dual Agent layer of the embodiment of the invention consists of a main Agent and a standard analysis Agent, wherein the main Agent and the standard analysis Agent cooperate in ReAct modes, namely, agent1 performs task scheduling, page acquisition, rule calling and report generation in detection execution, when false report, missing report or rule conflict is found, exception and feedback information are transmitted to Agent2, agent2 performs reasoning based on a feedback result, recognizes conflict or deficiency of a standard item and a rule, takes action to correct the rule, optimize a script or release a new version, and finally transmits the result to Agent1 for execution. Thus, a cross-Agent 'reasoning-action-feedback-re-reasoning-re-action' loop is formed, and the continuous evolution of rules and execution is realized.

In the concrete implementation, the Agent1 mainly relies on deterministic flow steps, and the MCP tool is called to realize operations such as user intention analysis, equipment session and page acquisition, RIT modeling and consistency check, sandbox discrimination script execution, discrimination basis and report generation and the like, so that the stability and traceability of a detection result are ensured, and the Agent2 internally focuses on rules and script management, and ensures the accuracy and long-term evolution of a rule system through links such as entry analysis, DSL rule generation and verification, script synthesis and test, conflict resolution and version management. In these internal flows, partial links (e.g., anomaly diagnostics, conflict resolution) embody local inference + action logic, but the whole is more biased towards the invocation and management of the engineering tool chain.

The double-layer design not only utilizes the feedback closed-loop advantage of ReAct modes among the intelligent agents, but also keeps the flow controllability of the inside of each intelligent agent, thereby realizing a high-efficiency automatic, reproducible, evolution-traceable compliance detection system in a link of 'management specification, specification entry, DSL rule, discrimination script, discrimination result, discrimination basis and detection report'. The main Agent (Agent 1) acts as a 'task orchestration and execution core' in the television page compliance detection system. The method is responsible for converting the detection intention input by a user into an executable task, driving the page collection and RIT to generate and update, calling the issued DSL rule and the discriminant script to execute in a sandbox running environment, and monitoring the execution process in real time and processing the abnormality. After the task is completed, the intelligent agent gathers and attributing the detection result, and calls the judging basis and the report generating module to output the standardized detection report which can be submitted and traced. Meanwhile, the Agent1 feeds back abnormal information such as false alarm, missing alarm, unresolvable selector or threshold conflict found in the running process to the Agent2, so that a loop of rule optimization and version iteration is formed. Agent1 takes tasks as centers and results as targets, and ensures the controllability, reproducibility and traceability of a detection link 'management specification- & gt specification entry- & gt DSL rule- & gt discrimination script- & gt discrimination result- & gt discrimination basis- & gt detection report'. The specific function design principle is as follows:

The function of detecting intention understanding and task orchestration in the embodiment of the invention refers to the function of converting detection requirements (target equipment, page range, associated specification entries, execution time windows and SLAs) input by a user into executable task units. The implementation principle comprises the steps of analyzing a target page (such as a home screen and a subscription portal), solidifying associated specification item numbers and version information, disassembling a detection target into a machine type-by-page operation grid according to television equipment information (model, resolution, region and system version), configuring concurrency, overtime and retry strategies, matching an acquisition instruction script with an issued rule version for each task fragment, generating early warning for the condition of missing or incompatibility, establishing a binding relation of tasks, sessions, RIT and the rule version, and generating a unique traceable task number (job_id).

The detection task execution function in the embodiment of the invention is that the function is that page navigation, snapshot acquisition and RIT modeling are completed on a real television terminal, and a judging script is executed in a sandbox running environment to produce violation judgment. The realization principle comprises the steps of calling session management and remote control signal simulation of television equipment, driving a terminal to enter a target page and triggering screen frame capture, calling EPG page identification and structure modeling to generate an RIT node tree, executing consistency check, supplementing integrity through increment updating and version snapshot if necessary, reading and assembling an adaptive rule set version (ruleset _id@version), calling a judging script compiling and sandbox operation environment preparation, unifying dependence and resource configuration to generate an operation environment number (env_id), applying the judging script to the whole RIT tree in the sandbox, positioning related nodes or paths according to rule selectors and conditions, completing logic judgment and threshold comparison, monitoring and outputting operation logs and indexes through a judging process, and calling a violation interface detection reproduction to generate a minimum reproduction scene and a comparison material if necessary.

The function of outputting the detection result in the embodiment of the invention is to aggregate the detection result and generate a traceable judgment basis and a detection report, and simultaneously feedback the abnormality to the Agent2. The implementation principle comprises the steps of aggregating violation results according to task, page and rule entry dimensions, calling a discrimination basis generation and storage tool function, cutting, marking and aligning RIT paths and measurement information in the violation areas to form a discrimination basis list (evaluation_list_id) and solidifying the discrimination basis list into a long-term traceable archive, calling a detection report to generate and store, outputting a PDF/XLSX standardized report with rule entry references, rule and script versions and rule term

links, and archiving the report standardized report to form a report number (report_id), and meanwhile, sorting false reports, missing reports, conflicts and executing anomalies to form an abnormal feedback list to be pushed to an Agent2 for rule conflict resolution and version optimization.

In the embodiment of the invention, a standard analysis Agent (Agent 2) is positioned as a rule analysis and optimization core in a television page compliance detection system. The method is responsible for converting management specification items issued by a broadcast television administrative department into DSL (Domain specific language) rules and discriminant scripts executable by a system, and ensuring the accuracy and stability of a rule system under different page structures and versions through automatic verification and conflict resolution. Agent2 is not directly oriented to the user, but is triggered and invoked by a main Agent (Agent 1), and a rule analysis flow is started when rule deletion, abnormality or conflict is found in the execution of a detection task. Through analysis of the specification items, generation of rules and verification and release of scripts, the Agent2 and the Agent 1 form a closed loop of 'task arrangement-rule generation-execution feedback-rule optimization', and the continuous availability, evolution and traceability of 'management specification, specification items, DSL rules, judgment of scripts, judgment results, judgment basis and detection report' of a detection link are ensured. The specific function design principle is as follows:

The rule analysis task scheduling function in the embodiment of the invention refers to preprocessing the submitted standard item set, version and abnormal context when the Agent 1 triggers to form a rule analysis task. The realization principle comprises the steps of calling a management specification document import and specification item segmentation to obtain an authoritative item list, calling a specification item vector generation and comparison tool to perform semantic search, positioning item contents related to a detection task, supplementing positive and negative case information by combining a violation type case base, and solidifying item numbers, versions and detection points (such as column number threshold and region proportion requirements) required by the task to form an analysis task unit. And finally, outputting a rule analysis task number, a specification entry index and a key point abstract as input for generating a follow-up rule.

The analysis task execution function in the embodiment of the invention is to convert the specification item into an executable rule and complete automatic verification of grammar, semantics and security level. The implementation principle comprises the steps of calling a DSL rule generating tool to generate a rule draft based on an item and a page type, calling DSL grammar checking and DSL semantic checking to ensure that a rule field is complete, a selector is resolvable, a threshold value is reasonable and consistent with a RIT structure, calling a tool to generate a judging script based on DSL and RIT to compile the rule into a script draft, eliminating override calling and non-compliance operation through script safety and specification checking, then calling a sandbox running environment to execute the script in a small scale, collecting accuracy, recall rate and boundary sample, comparing false alarm/missing report fed back by Agent 1 with a non-resolvable selector list, triggering conflict resolution logic (coverage conflict, repetition conflict and priority conflict), and correcting the rule through merging, splitting or priority ordering. And finally outputting the corrected rule draft and conflict resolution report.

The function of judging script output in the embodiment of the invention is to pack and issue the rule and the script after verification and conflict resolution, and to incorporate versioning management and knowledge precipitation. The realization principle comprises the steps of calling a rule release tool to generate a unique number and version number (ruleset_id@version) for a rule set, designating an applicable page and an effective range, calling a rule version management tool to realize version reading, difference comparison and rollback, and simultaneously precipitating the current generation and conflict resolution process into DSL templates, reverse modes and item mapping links for subsequent rapid generation and optimization multiplexing. Finally, the rule set number and version number, the published DSL rule text and the discriminant script source code are output, the template and the reverse mode library are updated, and the template and the reverse mode library are returned to the Agent 1 for calling and executing.

In specific implementation, the visualization layer of the embodiment of the invention plays a role of unifying entry of man-machine interaction and information presentation in a television page compliance detection system. The method connects a front-end user with a background agent and MCP service, and guarantees a complete closed loop from task creation to result presentation in the detection process. The design of the method follows the core targets of 'user friendliness, controllable process and traceable result', and ensures the transparency, compliance and traceability of detection work through tasking configuration, procedural monitoring and resultants display. The module is wholly divided into five functions including user entrance, inspection process control, television standard management, detection result and basis presentation and system management, and the five parts are mutually independent and tightly connected, so that standard operation of a full chain of compliance detection is ensured.

In the embodiment of the invention, the user entry function refers to the capability of providing unified access and task configuration of the detection system. The function completes authority authentication and hierarchical management of different roles (such as an administrator, an inspector, an auditor or a supervisor) through the login and identity verification module, and ensures the safety and compliance of detection operation. The user can initiate a television compliance detection task at a task creation and configuration interface, fill in or import vendor model, terminal platform, region and firmware version information, and select associated specification items (e.g. "first page first screen free special area is greater than or equal to 50%" "navigation column number is less than or equal to 15"). The system generates a unique number for each task and binds with the back-end detection session to form a unified identification of the whole process of the task.

In the embodiment of the invention, the checking process control function is to provide real-time control and visual monitoring on the task execution state for a user. The function supports the starting, suspending, stopping and retrying operation of tasks through a television system detection console, and simultaneously accesses the EPG page acquisition stream in real time to display the page snapshot and the operation log. By means of the real-time monitoring task monitoring module, a user can visually check the progress and the running condition of a task link, and the detection process is transparent, controllable and traceable from page acquisition, RIT generation, rule execution and judgment basis generation through progress bar, log pushing and running index display.

The television specification management function in the embodiment of the invention refers to unified display and visual maintenance of management specifications and specification entries. The user can search, browse and compare the broadcast and television standard item library on the front-end interface, wherein the item library comprises information such as item text, version number, effective period, application range and the like, and the information is displayed in linkage with the mapping relation of DSL rules and judging scripts. The user can also check the version evolution condition of the item and support the traceable link from the specification item to the rule and the script. The front-end system is linked with the standard search service, so that authority and traceability of the item are guaranteed, a rule simulation interface is provided, a user can conveniently and immediately run a rule set on a designated RIT page, and the executing effect of the item to the rule is intuitively verified.

In the embodiment of the invention, the detection result and basis presentation function refers to visual display of the detection output violation result and the judgment basis. A user can screen violation items according to the standard items, pages or time dimensions through a result browser, and each violation record can be unfolded to view the page snapshot, the cutting and labeling graph, the RIT path and the corresponding discrimination logic. The time line playback function supports the whole process of reproduction tasks, rule execution is acquired from pages, consistency of user verification processes and results is facilitated, and the comparison analysis function supports differentiated display of detection results of different versions or different devices. The user can also generate a PDF/XLSX standardized report through one key of the automatic report generator, and download the PDF/XLSX standardized report according to the packaged file (screenshot, label drawing, RIT slice, metric value and verification fingerprint), so as to ensure the transferability and long-term traceability of the detection result.

The system management function in the embodiment of the invention refers to unified configuration and audit of the operation, tasks and users of the detection platform. The function comprises task and history archiving, supporting retrieval and archiving based on manufacturers, models, versions or detection time to form a complete task archive link, user and operation log management, recording all key operation tracks (task running, rule modification, report generation and downloading), and guaranteeing auditability and compliance of a detection process. Through the functions, the system can meet the double requirements of the supervision department and the enterprise internal examination, and ensure the safety and stability of the detection platform in long-term operation.

In specific implementation, the internet television system model identification method based on the resource identification tree comprises the following steps:

S1, data modeling is used as a basic technical link for realizing the algorithm, and the core is to construct a priori knowledge model-resource identification tree for defining hierarchical logic relationship of elements of an Internet television interface, construct a data set with visual characteristics and semantic association dual attributes based on the tree model, and provide standardized support for model training and reasoning processes. Aiming at the interface hierarchy characteristics of the Internet television system, a system-page-column-element resource identification tree is constructed, and the tree structure and each hierarchy are defined as follows:

Referring to fig. 8, the tree structure of the embodiment of the invention adopts a multi-tree data structure, the root node is the whole internet television system, the first-level child node corresponds to a "page", the second-level child node corresponds to a "column" under the page, the third-level child node corresponds to an "element" under the column, and each node comprises a unique identification ID, an attribute set, a father node pointer and a child node list 4-type core field. The page level (primary node) is defined as an independent functional interface in the Internet television system, such as a first page, a video channel page, a free special area page and the like, and the special pages such as the first page, the free special area, the application page and the like are marked according to related standards. The column level (secondary node) is defined as a module with an explicit functional partition in the page, and covers the core types of navigation columns (such as top "home page-movie-variety" navigation), application columns (such as side "game-education-music" application list), recommendation areas (such as home page "hot recommendation" and "recommending for you" content plates), search columns, setting item areas and the like, so that the association and functional distinction of the column and the page are realized. Element level (three-level node) is defined as the minimum interaction or display unit in the column, and comprises picture elements (such as a recommended area poster and an application column icon), text elements (such as navigation column characters and a recommended area title), button elements (such as a play button and a download button) and input frame elements 4 types, so that basic data support is provided for a follow-up interaction algorithm.

Further, in the embodiment of the invention, the internet television specification is intended to analyze and compliance detection domain specific description language (DSL) generation method. As shown in fig. 9, in the rule semantic analysis stage, a multi-stage analysis framework based on deep learning is adopted, entity and relation extraction is performed on rule texts by using a Legal-BERT model, and key elements such as obligation subjects, behavior conditions, constraint terms and the like are identified. Aiming at the special terms in the television field, the system constructs a broadcast and television field ontology library containing 872 professional terms, solves the word ambiguity problem through a semantic disambiguation technology, and finally outputs an Abstract Syntax Tree (AST) in a JSON Schema format, wherein each node contains a term type, a logic operator, a conditional expression tree and an associated rule original text position identifier. This structured representation enables the subsequent processing to accurately understand the semantic requirements and logical relationships of the regulations. As shown in fig. 10, in the policy DSL generation stage, interface element data provided by the AST and UI recognition module output by the semantic analysis module is used as dual input. The system first performs template matching and selects a base framework from a library of preset templates according to the type of terms in the AST (e.g., price terms or path terms). Specific parameters are then injected into the template, which process entails dynamically binding UI element properties and automatically calculating click path depth from its parent container hierarchy. The finally generated DSL strategy is subjected to double verification, namely, whether the grammar accords with the domain language specification is checked, and whether all the referenced UI attributes exist is verified, so that the output strategy can be directly compiled and executed.

In the invention, the television pages with complex and various structures are converted into a hierarchical structured model by constructing a television page resource identification tree, the steps of controlled acquisition of the television pages, structured modeling of page information, generation and management of compliance discrimination rules, sandbox execution, detection report generation and the like are abstracted into services which can be called by MCP protocol, and automatic arrangement of detection tasks is realized through cooperation of an intelligent agent and a standard analysis intelligent agent, so that a closed-loop system from task configuration, page acquisition, RIT modeling, rule generation, discrimination execution to report archiving is established, and the problems that the prior art cannot adapt to supervision requirements of complex and various television pages and dynamic update are effectively solved, the full-flow automation and dynamic adaptation traceability of a compliance detection chain is improved, and the detection result can be improved.

In addition, the embodiment of the invention abstracts the screen shot of the television EPG page into a layering structure of page, column and element for the first time, and introduces the attributes of element type, coordinates, area proportion, saliency and the like to construct a computable structured model. The method breaks through the limitation that HTML/DOM analysis cannot be used in a television environment, is directly oriented to an actual presentation interface, ensures the stability and consistency of detection input, and provides a solid foundation for rule execution and index measurement.

The invention converts the management specification entry into the executable DSL rule and the discrimination script through the specification analysis agent, and introduces grammar and semantic verification, conflict resolution, regression verification and versioning management mechanisms to form a traceable link from 'specification entry- & gt DSL rule- & gt discrimination script- & gt discrimination basis'. The control mechanism not only solves the problem that the rules are difficult to maintain and lack consistency in the existing method, but also ensures that detection tasks under different time and different version specifications can be reproduced.

Furthermore, the invention introduces a sandbox running environment, provides unified dependence, resource quota and random seed configuration for the discrimination script, and ensures that consistent output is obtained under the same input. The running process records logs, performance indexes and illegal page reproduction scenes, supports comparison and rechecking of cross-version and cross-equipment, and remarkably improves the stability and the interpretability of detection. In the detection result output link, the invention correspondingly combines the violation result with the page snapshot, the RIT path and the numerical measurement to generate a judgment basis and a standardized report with label and measurement information. Compared with the traditional mode of only outputting screenshot or abstract log, the report of the invention is more transparent and visual, and is convenient for the supervision and the understanding and verification of the supervision and the manufacturer and the user.

The invention constructs a closed loop system from task configuration, page acquisition, RIT modeling, rule generation, discrimination execution to report archiving. All detection products (task numbers, television equipment information, page snapshots, RIT models, rule versions, discrimination basis and detection reports) keep a binding relation, support tracing and reproduction of cross tasks and cross time periods, and make up for the defects of task fragmentation and non-traceability of the process in the existing scheme.

The embodiment of the invention also provides a television page compliance detection system, which comprises a compliance detection main intelligent agent and a specification analysis intelligent agent, wherein the compliance detection main intelligent agent is used for controlling the establishment of a task according to television page detection requirements and determining management specification items corresponding to the task, triggering the acquisition of data of a television page to be detected, extracting the television page to be detected into a hierarchical structure of a page-column-element by utilizing the acquired data, adding attributes into the extracted hierarchical structure and generating a resource identification tree RIT (RIT), converting the television page to be detected into a computable structural model, triggering the execution of a discrimination script on each resource identification tree of the structural model in a sandbox operation

environment, outputting a violation result, an operation log and a performance index, and carrying out the reproduction of the violation page according to the triggering, combining the violation conclusion with corresponding rule page snapshot, RIT paths and numerical measures, generating a discrimination basis with label and measurement information, and generating a standardization report, and returning the analysis intelligent agent to the management specification, converting the management specification items into a computable structural model, carrying out discrimination script, carrying out the discrimination script and verification and the performance.

Meanwhile, the embodiment of the invention also provides a computer readable storage medium, wherein a computer program is stored on the storage medium, and when the program is executed by a processor, the method for deducing the artificial intelligence innovation ecological construction strategy based on the multi-agent game is realized.

The relevant content of the system embodiments and the storage medium embodiments of the present invention may be understood with reference to the method embodiments of the present invention, and will not be discussed in detail herein.

Although the preferred embodiments of the present invention have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, and accordingly the scope of the invention is not limited to the embodiments described above.

Document Claims

Language: (Chinese)

1.一种电视页面合规性检测的方法，其特征在于，所述方法包括：

根据电视页面检测需求创建任务，并确定该任务所对应的管理规范条目；

采集待检测的电视页面，对采集到的电视页面进行结构化解析与元素信息识别，将页面信息抽取为“页面—栏目—元素”结构的层次化、可计算的数据模型，即资源信息树RIT；

将所述管理规范条目转化为可执行的判别脚本，并对转换后判别脚本进行语法与语义校验、冲突消解、回归验证和版本化管理；

在沙箱运行环境下，对所述结构化模型的每个资源标识树执行判别脚本，输出违规结果、运行日志和性能指标，并根据触发进行违规页面复现；

将违规结果与对应的违规页面快照、资源标识树路径和数值度量相结合，生成带有标注与度量信息的判别依据，并生成标准化报告。

2.根据权利要求1所述的方法，其特征在于，所述根据电视页面检测需求创建任务，并确定该任务所对应的管理规范条目，包括：

根据电视页面检测需求创建任务，固化任务上下文并生成任务编号，所述任务编号用于唯一标识任务，所述任务中包括并发、超时、重试策略与权限域；确定所述任务所对应的管理规范条目，并获取管理规范条目的编号、版本、生效期与锚点，将所获取的管理规范条目与所述任务进行绑定。

3.根据权利要求1所述的方法，其特征在于，将所述管理规范条目转化为可执行的判别脚本，并对转换后判别脚本进行语法与语义校验、冲突消解、回归验证和版本化管理，包括：

通过规范解析智能体将所述管理规范条目转化为可执行的DSL规则，并将所述DSL规则转换为判别脚本，对转换后的DSL规则与判别脚本进行语法与语义校验、冲突消解、回归验证和版本化管理。

4.根据权利要求3所述的方法，其特征在于，将所述管理规范条目转化为可执行的DSL规则，并将所述DSL规则转换为判别脚本，还包括：

当数据库中存在所需的管理规范条目、DSL规则和判别脚本，则直接调用；

当数据库中缺失所需的管理规范条目、DSL规则和判别脚本，则通过所述规范解析智能体生成对应的管理规范条目、DSL规则和判别脚本，并对所生成的管理规范条目、DSL规则和判别脚本进行语法和语义校验、安全与规范检查后存储在数据库中。

5.根据权利要求1所述的方法，其特征在于，

为判别脚本构建隔离运行环境，统一依赖、资源和随机种子，并生成运行环境编号，其中，所述运行环境编号用于标识沙箱执行实例；

在沙箱运行环境下，对整个结构化模型依据DSL规则中的选择器和判定条件定位相关的资源标识树节点或路径，执行与管理规范条目对应的逻辑运算与阈值比对，并输出违规结果、运行日志和性能指标；

对检测结果进行聚合和整理，按照页面与管理规范条目维度组织违规判别结论，并标注异常样本与边界情况，为后续反馈与DSL规则优化提供参考。

6.根据权利要求1-5中任意一项所述的方法，其特征在于，所述采集待检测的电视页面，包括：

与目标电视设备建立受控会话，采集页面快照和关键帧。

7.根据权利要求6所述的方法，其特征在于，将待检测的电视页面转换为层次化、可计算的结构化数据模型，包括：

对采集的页面快照和关键帧进行元素识别和结构建模，抽取为“页面—栏目—元素”的层次化结构，对抽取后的层次化结构中加入属性，以实现将待检测的电视页面转换为可计算的数据模型，其中，所述数据模型为带有路径和度量信息的资源标识树，所述属性包括元素类型、展示信息、坐标和面积比例；

生成资源标识树编号，对资源标识树进行一致性检查，并对层次化结构不一致的资源标识树节点进行修正。

8.根据权利要求1-5中任意一项所述的方法，其特征在于，生成标准化报告之后，所述方法还包括：

根据标准化报告中的误报、漏报和冲突情况来调整所述规范解析智能体，以提高所述规范解析智能体匹配识别准确性。

9.一种电视页面合规性检测系统，其特征在于，该系统包括：合规检测主智能体和规范解析智能体，其中，

所述合规检测主智能体，用于控制根据电视页面检测需求创建任务，并确定该任务所对应的管理规范条目；触发采集待检测的电视页面的数据，利用所采集的数据将待检测的电视页面抽取为页面—栏目—元素的层次化结构，对抽取后的层次化结构中加入属性，并生成资源标识树RIT，以将待检测的电视页面转换为可计算的结构化模型；并触发在沙箱运行环境下，对所述结构化模型的每个资源标识树执行判别脚本，输出违规结果、运行日志和性能指标，并根据触发进行违规页面复现；将违规结论与对应的违规页面快照、RIT路径和数值度量相结合，生成带有标注与度量信息的判别依据，并生成标准化报告；

所述规范解析智能体，用于将所述管理规范条目转化为可执行的判别脚本，并对转换后判别脚本进行语法与语义校验、冲突消解、回归验证和版本化管理，并进行电视页面合规性验证和回归评估。

10.一种计算机可读存储介质，所述存储介质上存储有计算机程序，所述程序被处理器执行时实现权利要求1-8中任意一项所述的电视页面合规性检测的方法。

Language: (English)

1. A method of television page compliance detection, the method comprising:

creating a task according to the television page detection requirement, and determining a management specification item corresponding to the task;

collecting a television page to be detected, carrying out structural analysis and element information identification on the collected television page, and extracting page information into a hierarchical and computable data model of a page-column-element structure, namely a resource information tree RIT;

Converting the management specification item into an executable judging script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script;

Executing a judging script on each resource identification tree of the structured model under the sandbox running environment, outputting a violation result, a running log and performance indexes, and carrying out violation page reproduction according to triggering;

combining the violation results with the corresponding violation page snapshots, the resource identification tree paths and the numerical metrics, generating a judgment basis with labels and metric information, and generating a standardized report.

2. The method of claim 1, wherein creating a task according to the tv page detection requirement and determining a management specification entry corresponding to the task comprises:

Creating a task according to the detection requirement of a television page, solidifying the context of the task and generating a task number, wherein the task number is used for uniquely identifying the task, the task comprises concurrence, overtime, retry strategies and authority domains, determining management specification items corresponding to the task, acquiring the number, version, effective period and anchor point of the management specification items, and binding the acquired management specification items with the task.

3. The method of claim 1, wherein converting the management specification entry into an executable discriminant script and performing grammar and semantic verification, conflict resolution, regression verification, and versioning management on the converted discriminant script comprises:

And converting the management specification entry into an executable DSL rule by a specification analysis agent, converting the DSL rule into a discrimination script, and carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted DSL rule and the discrimination script.

4. The method of claim 3, wherein translating the management specification entry into an executable DSL rule and translating the DSL rule into a discriminant script, further comprising:

when required management specification items, DSL rules and judging scripts exist in the database, the method is directly called;

when the required management specification items, DSL rules and judging scripts are deleted from the database, corresponding management specification items, DSL rules and judging scripts are generated through the specification analysis agent, grammar and semantic verification and security and specification inspection are carried out on the generated management specification items, DSL rules and judging scripts, and the management specification items, DSL rules and judging scripts are stored in the database.

5. The method of claim 1, wherein the step of determining the position of the substrate comprises,

Establishing an isolated running environment for judging the script, unifying dependence, resources and random seeds, and generating a running environment number, wherein the running environment number is used for identifying a sandbox execution instance;

in a sandbox operation environment, the whole structured model locates relevant resource identification tree nodes or paths according to selectors and judgment conditions in DSL rules, performs logic operation corresponding to management specification items and threshold comparison, and outputs violation results, operation logs and performance indexes;

And (3) aggregating and sorting the detection results, organizing rule violation discrimination conclusions according to the dimensions of the page and the management specification item, marking abnormal samples and boundary conditions, and providing references for subsequent feedback and DSL rule optimization.

6. The method according to any one of claims 1-5, wherein the capturing the tv page to be detected comprises:

and establishing a controlled session with the target television equipment, and collecting page snapshots and key frames.

7. The method of claim 6, wherein converting the tv page to be detected into a hierarchical, computable structured data model comprises:

Performing element identification and structural modeling on the collected page snapshot and key frames, extracting a hierarchical structure of 'page-column-element', and adding attributes into the extracted hierarchical structure to realize conversion of a television page to be detected into a computable data model, wherein the data model is a resource identification tree with path and measurement information, and the attributes comprise element types, display information, coordinates and area proportions;

Generating a resource identification tree number, carrying out consistency check on the resource identification tree, and correcting the resource identification tree nodes with inconsistent hierarchical structures.

8. The method of any of claims 1-5, wherein after generating the standardized report, the method further comprises:

And adjusting the standard analysis agent according to the false report, the missing report and the conflict conditions in the standardized report so as to improve the matching recognition accuracy of the standard analysis agent.

9. A television page compliance detection system is characterized by comprising a compliance detection main agent and a specification analysis agent, wherein,

The method comprises the steps of acquiring data of a television page to be detected, extracting the television page to be detected into a hierarchical structure of page-column-element by utilizing the acquired data, adding attributes into the extracted hierarchical structure, generating a resource identification tree RIT to convert the television page to be detected into a computable structural model, triggering the execution of a discrimination script on each resource identification tree of the structural model under a sandbox running environment, outputting a violation result, a running log and a performance index, and carrying out violation page reproduction according to the triggering, combining the violation conclusion with a corresponding violation page snapshot, a RIT path and a numerical measurement, generating a discrimination basis with labeling and measurement information, and generating a standardized report;

The specification analysis agent is used for converting the management specification item into an executable judging script, carrying out grammar and semantic verification, conflict resolution, regression verification and versioning management on the converted judging script, and carrying out television page compliance verification and regression evaluation.

10. A computer readable storage medium having stored thereon a computer program which when executed by a processor implements the method of television page compliance detection of any of claims 1-8.